

Quasi-Spherical Absorbing Receiver Model of Glioblastoma Cells For Exosome-based Molecular Communications

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ABSTRACT

In this paper, we propose a mathematical and computational model for the GBM initiating of recurring focus as a quasi-spherical absorbing receiver considering the irregular shape as a Bernoulli trial process that accounts for the uncontrollable tumor growth over an initial spherical surface. Our results show that when GBM grow to irregular quasi-sphere shapes, they will increase the channel capacity, which is fully aligned with the evolution and configuration of GSC niches in GBM cultures.

CCS CONCEPTS

• **Mathematics of computing** → **Information theory**; • **Applied computing** → **Systems biology**; • **Computing methodologies** → **Modeling and simulation**.

KEYWORDS

Mathematical Modeling, Cancer Targeted Drug-Delivery, Molecular Communications via Diffusion, 3D Modeling, Glioblastoma Multiform.

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1 INTRODUCTION

Molecular communications (MC) has been fast progressing as an interdisciplinary discipline from engineering to biotechnology and medicine, as a source of novel inspired technology to address complex diseases and health challenges. One great example, is the application of MC principles to the theranostics (therapy and diagnostics)

of Glioblastoma Multiforme (GBM), as pioneered by the EU-FET Gladiator project (www.fet-gladiator.eu). GBM is considered one of the most aggressive forms of brain tumors, with very limited advancements over the past thirty years. This is greatly attributed to aberrant vascularization, enabling fast tumor growth and spreading of glioblastoma stem cells (GSCs) which survive the current therapeutic schemes, leading to recurrence at or near the resection cavity which are the areas mostly irradiated [4]. MC can be used to investigate how the therapeutic molecules can be transported to GBM, and this can be through techniques that includes engineered release rates, signal modulation, coding to maximize the treatment efficacy as well as in novel solutions of externally controlled drug-delivery through wireless communications [1]. In this context, MC serves as a powerful tool in a wide range of theranostics applications, epitomized by cancer-targeted drug-delivery systems.

In this work we propose the modelling of a GBM as a quasi-spherical and absorbing receiver for exosome-based MC from induced Neural Stem Cells (iNSCs). We consider the hitting probability of exosomes in the GBM surface that is composed of semi-spherical and wedged shaped structures (the wedged shape is based on connections of spherical bubbles on the surface of the GBM) or also the original surface of the GBM that remained without semi-spheres.

2 QUASI-SPHERICAL ABSORBING RECEIVER MODEL OF A GBM

To model the GBM as molecular receivers, we have to take into account the irregularities on the cell surface, endowed by its invasive potential. Due to the uncontrollable growth and the continuous interplay between repopulating and invasive phenotypes, i.e. GSCs and GCs, the evolving cell population around the tumor surface over time, the initial spherical structure is transformed into a quasi-spherical shape. We assume that the GBM cells will be a combination of many semi-spheres plus wedges or holes at the surface which is in line with the observations in two-dimensional in vitro cultures of GBM cell lines. We consider the methods developed by Schulten and Kosztin [3] and Yilmaz et al. [5] to develop a model that obtains the fraction number of exosomes hitting the receiver as well as the expected number of hits at the receiver and then modify it from a spherical absorbing receiver to consider an irregular shaped GBM or a quasi-spherical model of the GBM. To account for molecules hitting the semi-spheres or the wedges of the GBM we use a Bernoulli trial, with probability of hitting the semi-spheres being defined as p , and the probability of hitting the

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GBM surface defined as $q = 1 - p$. This integrated model allow us to account for the presence of semi-spheres and wedges on the surface of the GBM cell due to the invasiveness and proliferation of malignant GCs. The modification lead to the new equations as follows,

$$N_{hit}(\Omega_r, t|d'_0) = \frac{r_r + r_s}{d'_0} \operatorname{erfc} \left(\frac{d'_0 - (r_r + r_s)}{\sqrt{4 \frac{D}{\lambda^2} t}} \right), \quad (1)$$

where N_{hit} represents the fraction of molecules absorbed, r_s represents the mean value for the radius of the semi-spheres on the surface of the GBM, d'_0 represents the sum of the distance between transmitter and receiver plus mean value for the radius of semi-spheres and the original radius of the GBM cell, or simply, $d'_0 = d + r_s + r_r$, D is the diffusion coefficient and λ represents the brain ECS tortuosity. The Eq. (1) describes the estimated fraction of molecules to be absorbed by the receiver with the semi-spheres on the surface. We also obtain the expected number of hits at the quasi-spherical model of the GBM receiver considering its composition of semi-spheres and its original surface not yet invaded by the GSCs. The equation for the expected number of hits is described as follows

$$E[N_{R_x}^{\Omega_r}(t, t + \Delta t)] = p \cdot N_{T_x}^{d'_0} [N_{hit}(\Omega_r, t + \Delta t|d'_0) - N_{hit}(\Omega_r, t|d'_0)] + (1 - p) \cdot N_{T_x}^{d_0} [N_{hit}(\Omega_r, t + \Delta t|d_0) - N_{hit}(\Omega_r, t|d_0)]. \quad (2)$$

The equation represents that for the total estimated number of hits, a percentage dependent on the value of p , will be calculated considering the semi-spheres, and a percentage of the total dependent on $(1 - p)$ will be calculated considering only the surface of the original GBM cell.

2.1 Mutual Information Analysis

To account for the amount of reliable information transfer between the iNSC and the GBM using exosomes, we calculate the channel capacity of this MC system and compare it to a spherical receiver GBM model. We use the conventional general representation of channel capacity to numerically account for maximum mutual information of the system basing on the model developed for diffusive channels including [2]. We consider just molecular release from the transmitter, simply one slot, $n = 1$, for the mutual information analysis and therefore, the channel capacity.

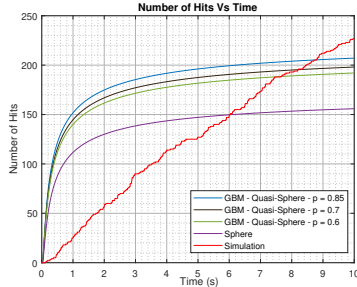


Figure 1: Number of molecular hits for GBM quasi-sphere and sphere model and the total number of molecules reacting to the receptors and absorbed by the receiver cell in the 3D Monte Carlo Simulation.

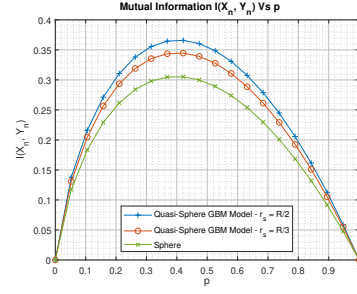


Figure 2: Mutual Information $I(X; Y)$ using the quasi-sphere GBM model compared to a perfect sphere as a receiver. The diffusion coefficient for the simulation was $D = 10 \mu m^2/s$.

3 RESULTS AND CONCLUSION

The Fig. 1 depicts the number of the molecular hits on the quasi-sphere GBM receiver model comparing it to a purely spherical model as well as a 3D Monte-Carlo simulation of molecular hits and absorption in a generated GBM model with semi-spheres on its surface in line with the mathematical model in this paper. From the Fig. 1 we can observe that the quasi-sphere GBM receiver model can better describe a more accurate representation of the GBM taking into account its physical features predicting more realistic results regarding molecular hits and absorption than when compared with a pure spherical representation of the GBM receiver. The Fig. 2 represents the mutual information for the quasi-spherical model of the GBM receiver with two different values for the semi-spheres' radius as well as for a sphere model of the GBM. From the results, we observe that the quasi-spherical model presents a higher mutual information, and consequently, channel capacity than a simple spherical model of the GBM. Our results suggest that the tumor can grow and develop faster due to the higher mutual information when compared with simpler spherical models of GBM tumor models. It can be further concluded that targeted drug-delivery systems could be more effective for the same reasons.

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